

Applying Fourier Transform to Approximate and Plot a Topographic Terrain

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Learning Primer

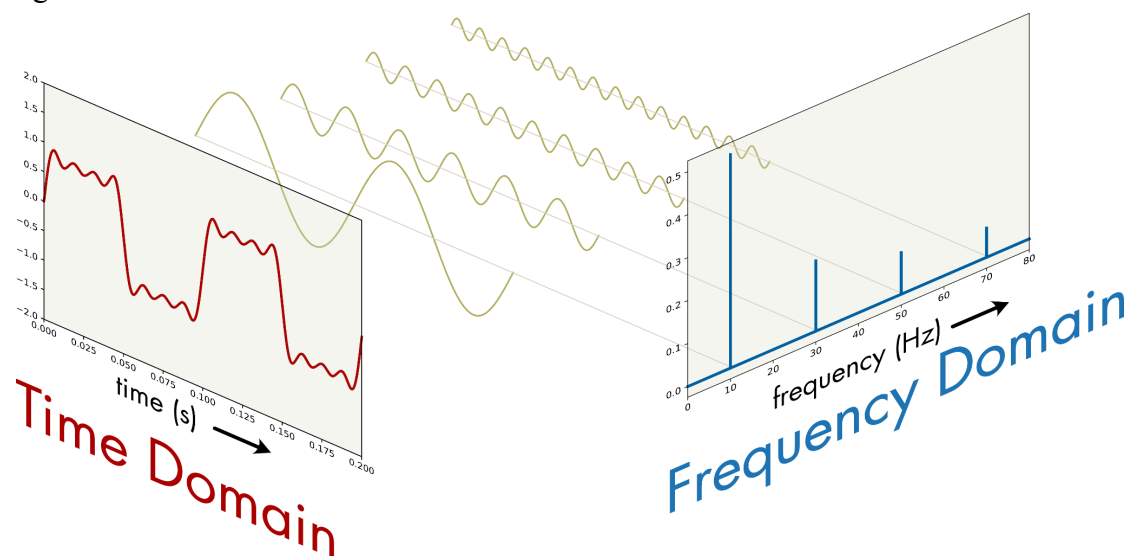
Learning Objective

Our main objective was to learn about and apply fourier transform to approximate and filter multivariable functions. Specifically, we wanted to explore how the radius of frequencies used in fourier transform makes the approximation of a topographic terrain more or less accurate.

Fourier Transform Summary

Fourier transform transforms a function in the time domain, $f(t)$, into a function in the frequency domain, $f(v)$, by decomposing the function into a sum of sine and cosine waves of specific frequencies. For our project, we utilized a *position* domain rather than a time domain since our $f(x, y)$ represents elevation changing with position (x, y) . When we consider a signal from our distance function, we slice our function horizontally or vertically to achieve a signal of elevation vs x or y position.

The time and frequency domains are pictured together in Figure 1. In the time domain, the signal's amplitude is plotted against time (or in our case, elevation vs position). In the frequency domain, the magnitudes of the sinusoidal waves are plotted against the frequencies of those waves, essentially showing how much of each frequency-value wave is present in the original signal.



(Figure 1: A function in the time domain vs in the frequency domain. For this example signal, the majority of frequencies that comprise it are within 10-30 Hz and the signal can be decently approximated by summing waves of these frequencies. However, adding greater-frequency waves to the sum makes the approximation increasingly accurate to the original signal. Image source:

https://neurotext.library.stonybrook.edu/C7/C7_1/C7_1.html)

Fourier Transform: Mathematical Description

The 2D fourier transform approximation of $f(x, y)$ over frequencies $[u, v] \in R$ is computed as:

$$f(x, y) \approx \sum_{u,v} C(u, v) e^{i(ux+vy)}$$

Rewritten using Euler's formula, this is:

$$f(x, y) \approx \sum_{u,v} C_{real}(u, v) \cos(ux + vy) - C_{imaginary}(u, v) \sin(ux + vy)$$

Where $\cos(ux + vy)$ and $\sin(ux + vy)$ represent 2D frequencies and the fourier coefficients C_{real} and $C_{imaginary}$ determine how much of each frequency comprises the signal. The fourier coefficients are solved for as follows:

$$C_{real} = \sum_{u,v} F(x, y) \cos(2\pi(\frac{ux}{N} + \frac{vy}{M}))$$

$$C_{imaginary} = \sum_{u,v} F(x, y) (-\sin(2\pi(\frac{ux}{N} + \frac{vy}{M})))$$

Where $F(x, y)$ is the inputted M -by- N grid of elevation points and $2\pi(\frac{ux}{N} + \frac{vy}{M})$ represents the phase of a signal. The fourier coefficients C_{real} and $C_{imaginary}$ represent the cosine and -sine projections of the distance image onto the signal frequency. This is because sinusoids are orthogonal, so multiplying $F(x, y)$ by a sinusoid cancels all frequencies to zero except for the frequency in question, leaving only the part of the image composed of that frequency.

When we limit the radius of (u, v) frequencies, we exclude higher-frequency signals from the sum. This can be written as:

$$\sqrt{u^2 + v^2} \leq radius$$

Decreasing the frequency radius makes our approximation of $f(x, y)$ less accurate but also less rough because we filter out noise caused by high frequencies. By tuning the radius of included frequencies, we can achieve an image of our preferred accuracy and roughness.

Interactive Activity

Our activity allows students to visualize fourier transform to approximate a live contour image, changing the frequency radius to increase or decrease the accuracy of the image approximation. We process distance data from a Kinect One sensor to create a contour plot, which can be edited via live keyboard input. This way, students can see the difference between images that are reconstructed using lesser and greater frequency radii.

Physical props in addition to the Kinect One include a projector and various objects suspended on a string in front of the sensor. The projector projects the contour image back onto the object for a real-life contour plot, so students can place objects in front of the sensor or even stand in front of it to see the contour plot projected onto the object or their body. Lastly, we display papers with core equations and descriptions, plus example plots.

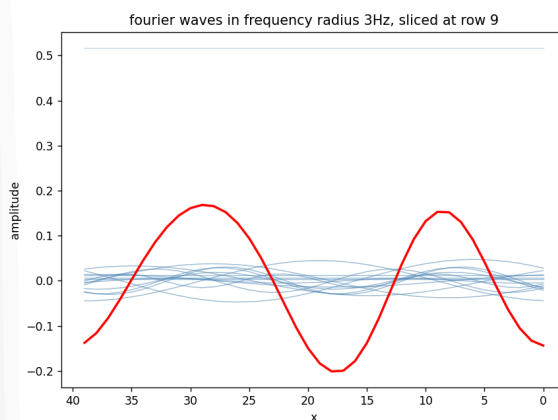
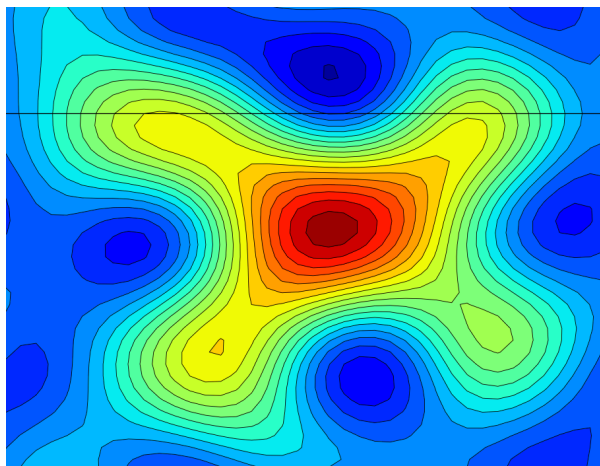
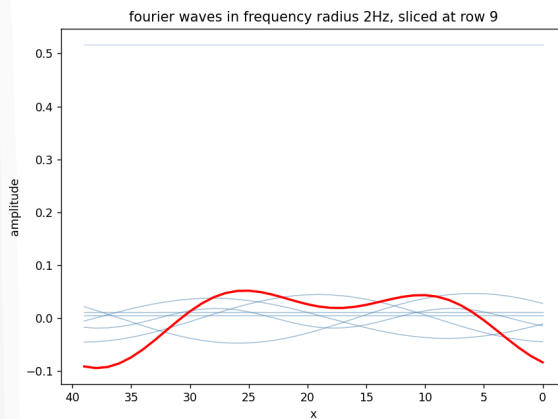
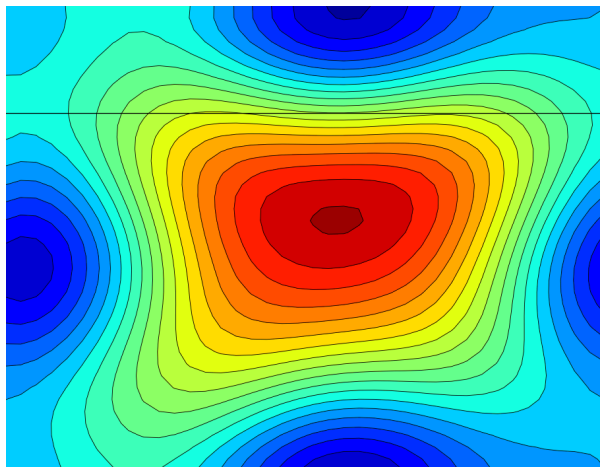
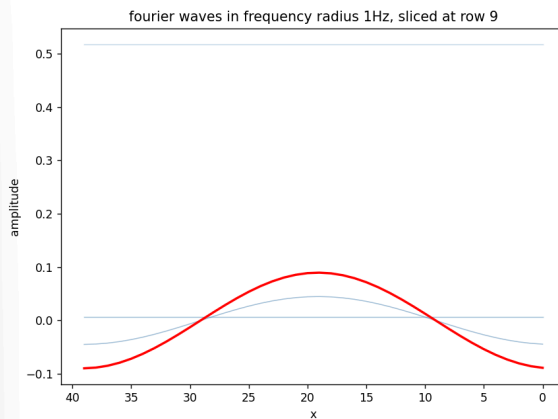
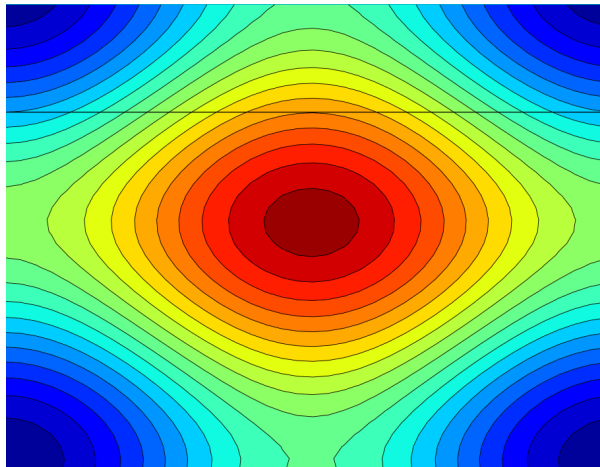
Students who tried our activity were: Liam, Will, Douglas, Akil, Max, Leo. Based on their feedback, our activity could be improved by a more detailed explanation of the math behind fourier transform.

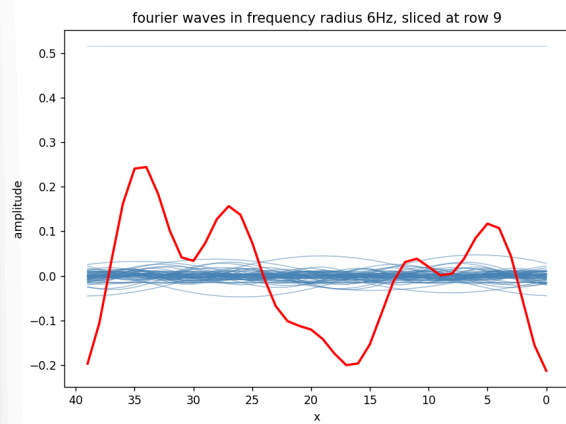
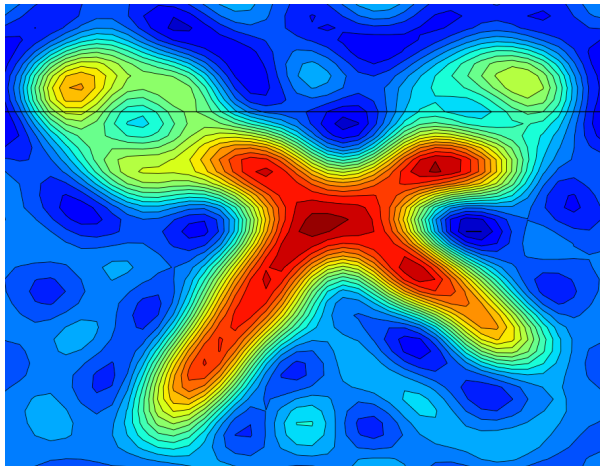
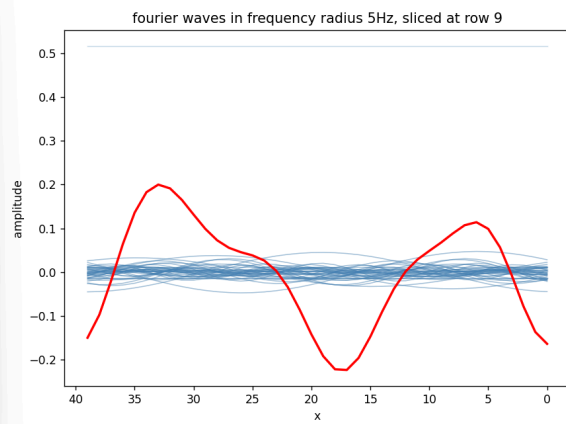
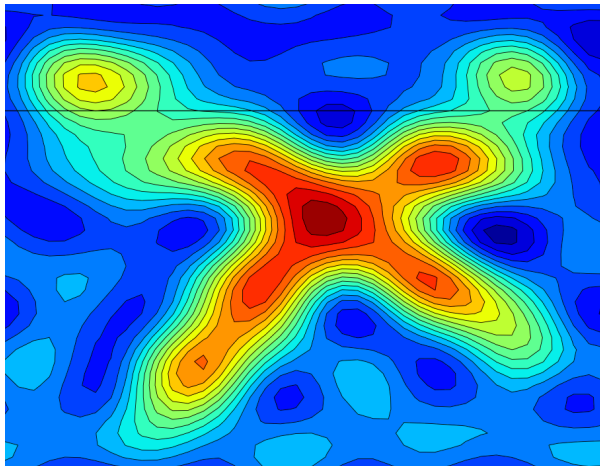
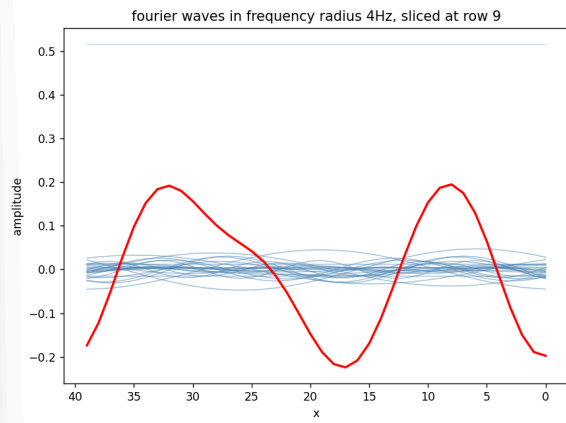
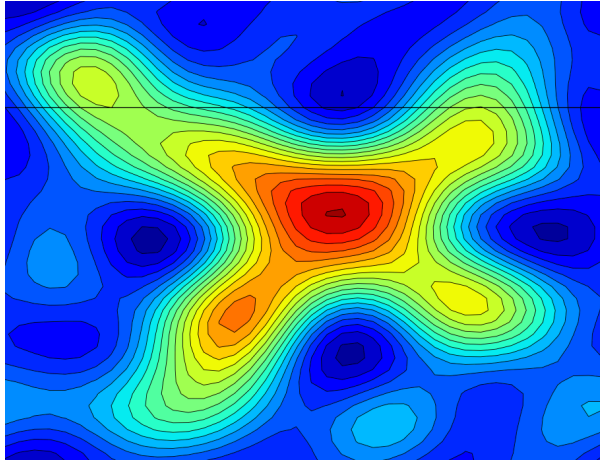


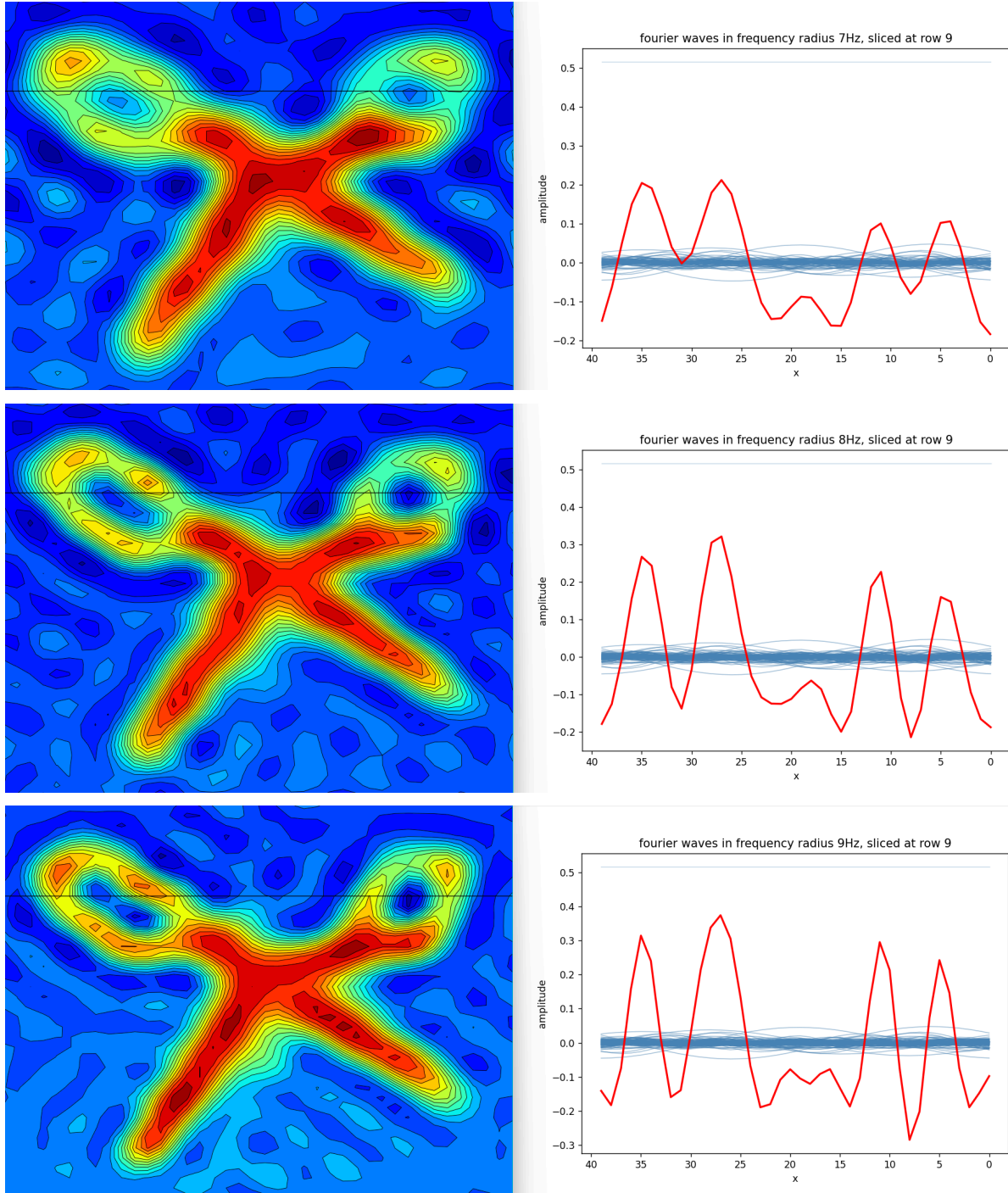
(Picture of our activity in the demo. We suspended an object in order to set it a significant distance away from the wall and create a dark blue background for our contour plot.)

Approximation of an Image Contour Using Fourier Transform

Below are nine side-by-side images we generated of an object's contour image and its fourier waves at frequency radii from 1Hz to 9Hz. As the radius increases, the sum of waves (right, red) and the image contour (left) become increasingly accurate to the real-world image.







(Figures 2.1-2.9: Increasingly accurate approximation of the contour image of a pair of scissors using fourier transform. As we increased the radius of frequencies used in our fourier transform function, our approximation became increasingly accurate and rough.)

Reference Materials:

MIT OpenCourseWare, Fourier Series

<https://www.youtube.com/watch?v=vA9dfINW4Rg&pp=ygUaZm91cmllciB0cmFuc2Zvcml0IG9wZW4%3D>

Medium, Fourier Transform 101

<https://medium.com/sho-jp/fourier-transform-101-part-1-b69ea3cb4837>

Khan Academy, Fourier Series

<https://www.khanacademy.org/science/electrical-engineering/ee-signals/ee-fourier-series>

Stonybrook, Signal Processing

https://neurotext.library.stonybrook.edu/C7/C7_1/C7_1.html

AI Disclaimer:

We utilized AI for aid learning about fourier transform, mainly for clarification and examples of the approximation process. We also used AI to debug our code and check errors which we could not resolve ourselves.